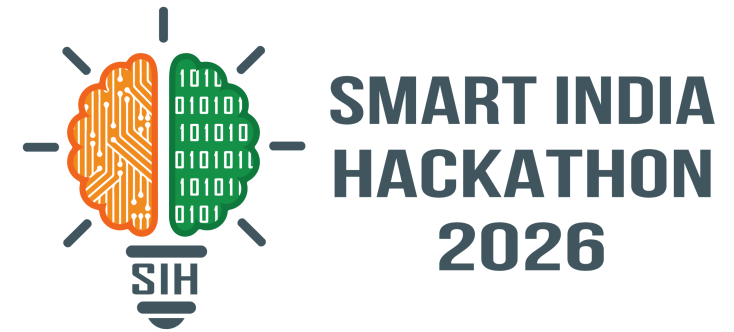


SMART INDIA HACKATHON 2026



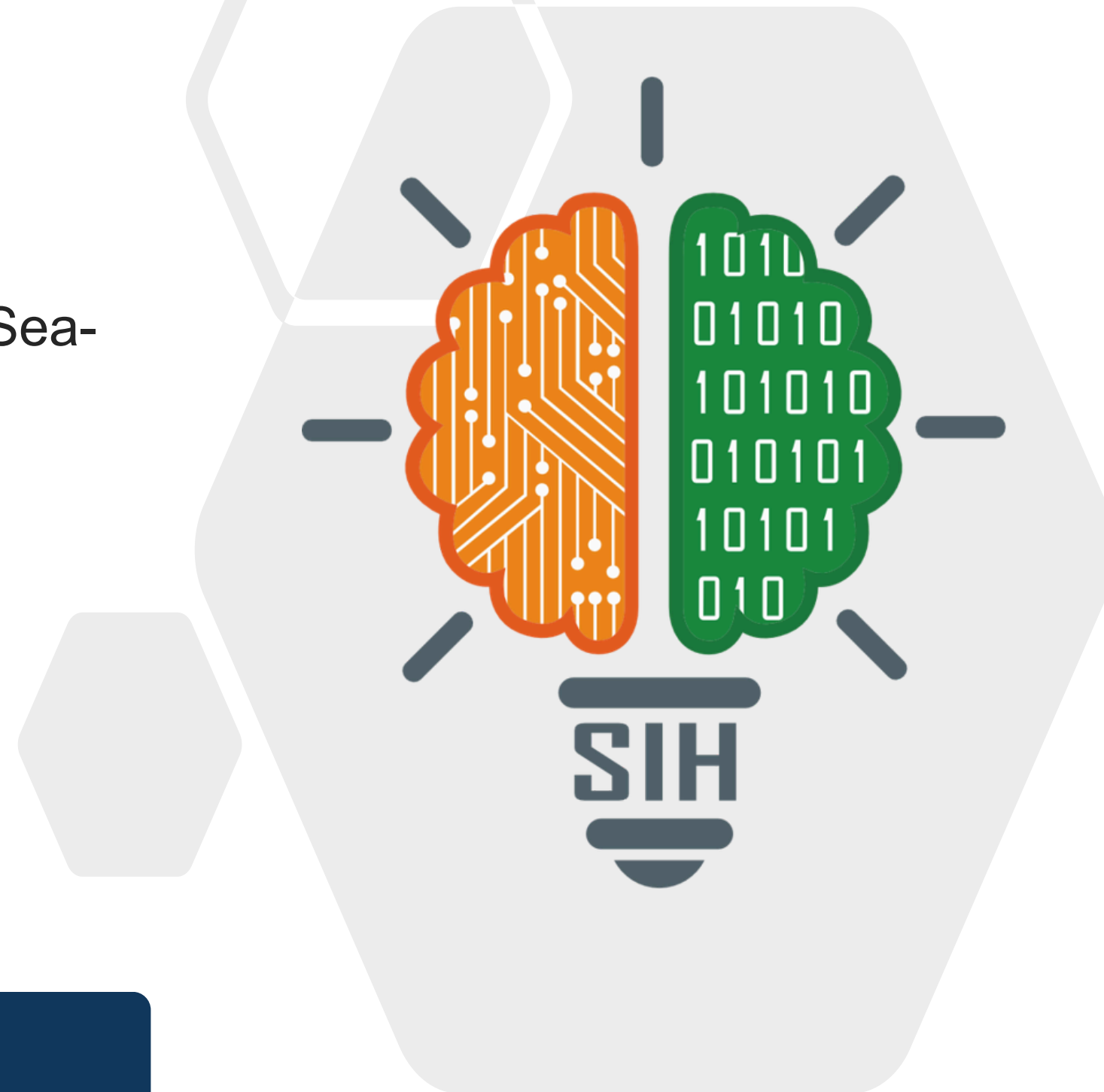
Problem Statement ID – 59

Problem Statement Title – AI-enabled Antarctic Sea-Ice, Iceberg Trajectory and Navigation Decision Support System

Theme – Smart Education

PS Category – Software

Team Name – 6OClock

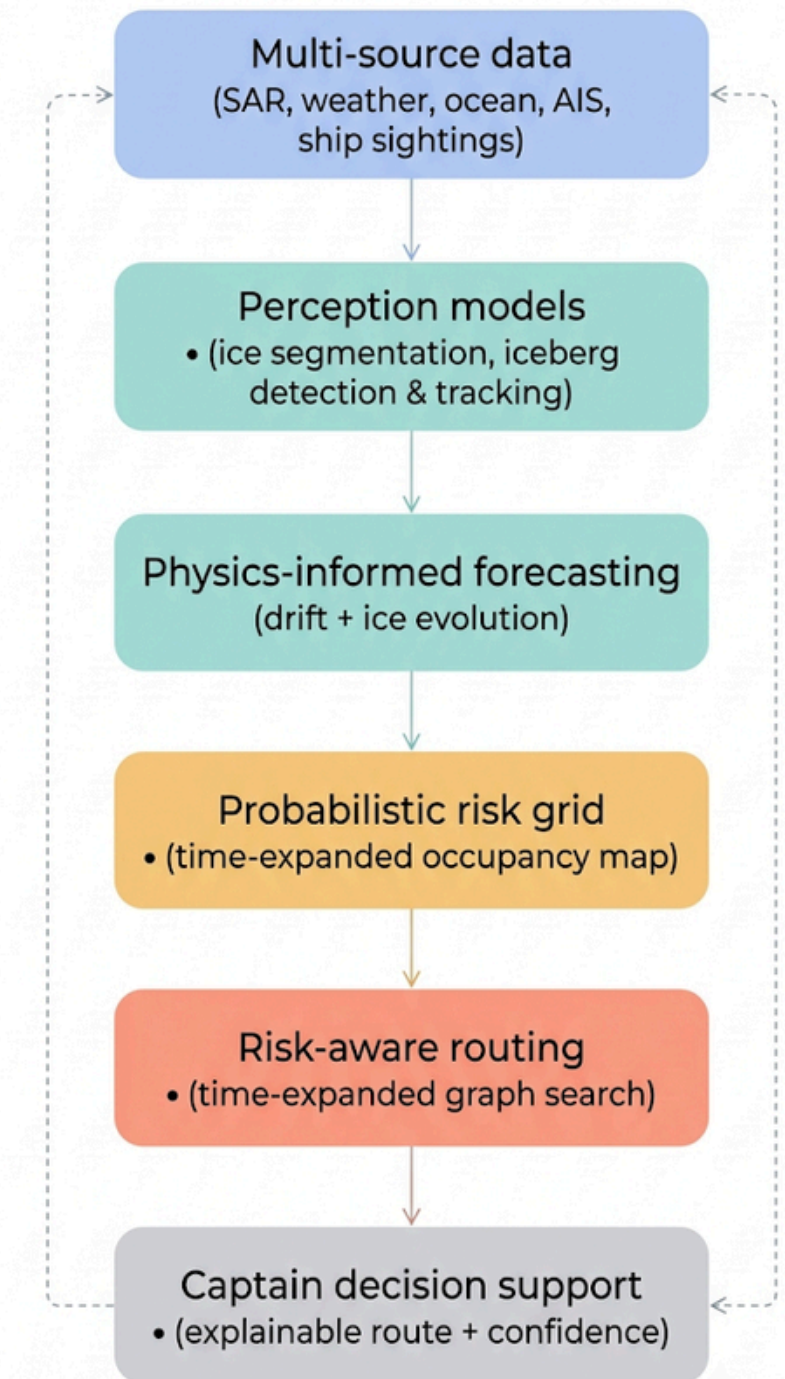
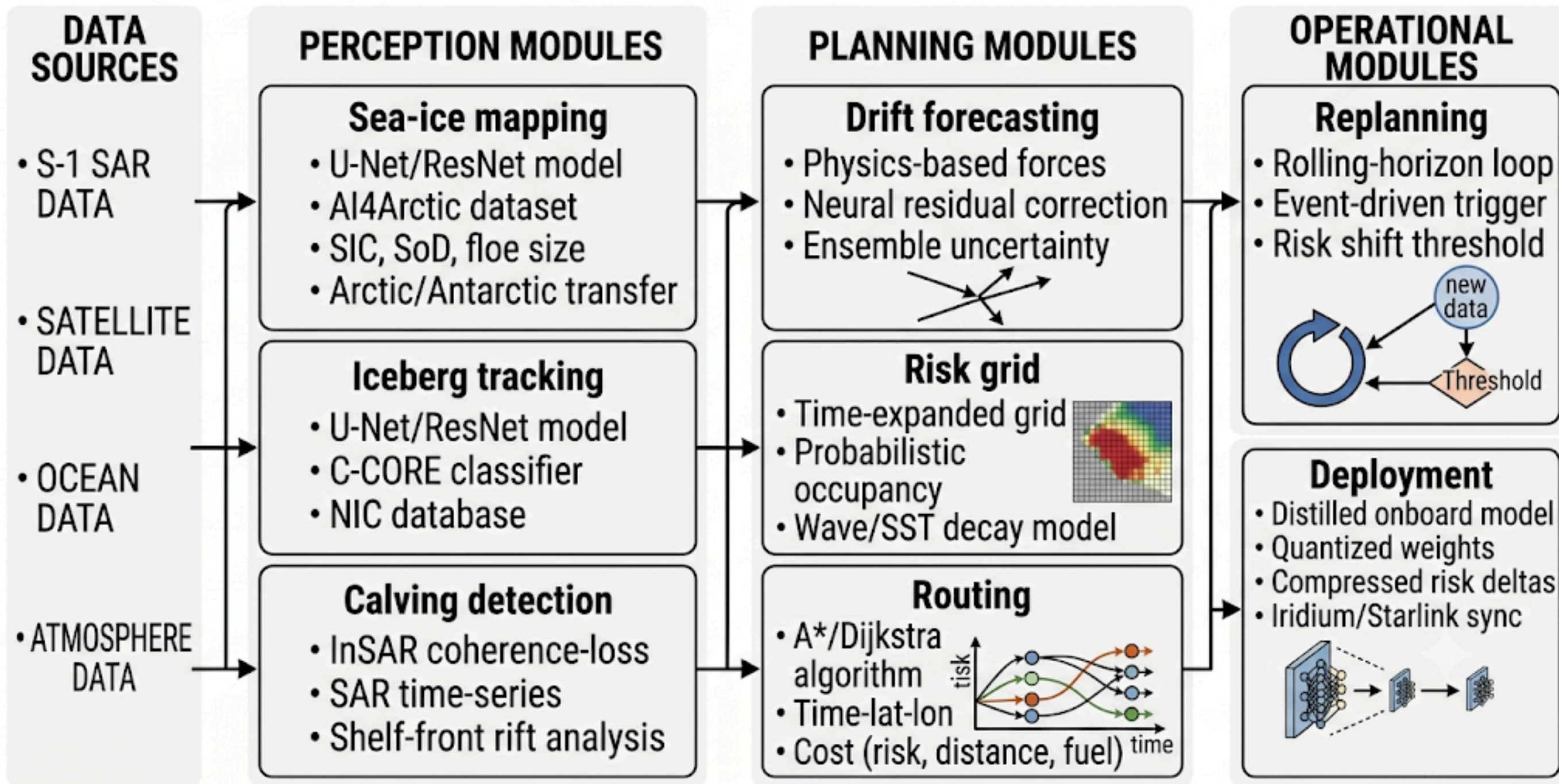


Integrated Smart Tourism Platform
Safety | Experience | Crowd Balance | Sustainability

AI-enabled Antarctic Sea-Ice, Iceberg Trajectory and Navigation Decision Support System

1. Where is the sea ice and icebergs going, how will conditions change, and what is the safest/most efficient route for a ship?
2. Satellite imagery + weather/ocean data + AI forecasting + navigation/risk analysis
3. Sea Ice changes continuously because of various ocean factors - where is the sea ice? what is its concentration? where will it form in two days?
4. Where is the iceberg? what is its trajectory? the path it is taking is mapped. Where will this iceberg be 6 hours, 12 hours, 24 hours, 3 days, etc. from now?
5. Risk map generation and route optimization using djikstra etc, because it has now become a graph problem

TECHNICAL APPROACH



Analysis of the feasibility of the idea

Technically feasible

builds on public datasets/models (AI4Arctic, Statoil/C-CORE, BYU/NIC) with transfer learning bridging the Arctic → Antarctic data gap.

Operationally feasible

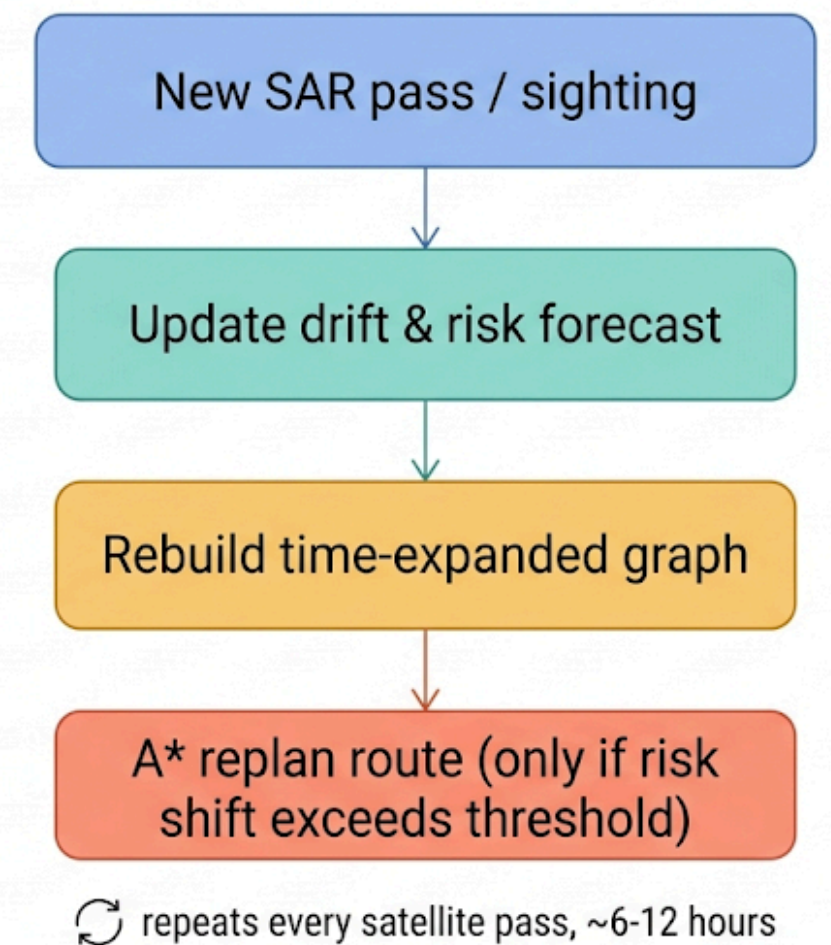
decision-support only : captain retains final authority; recommendations are explainable, not a black box.

Economically viable

models trained once, retrained infrequently (ocean/ice dynamics change slowly); no new sensors required , reuses existing satellite/SAR infrastructure.

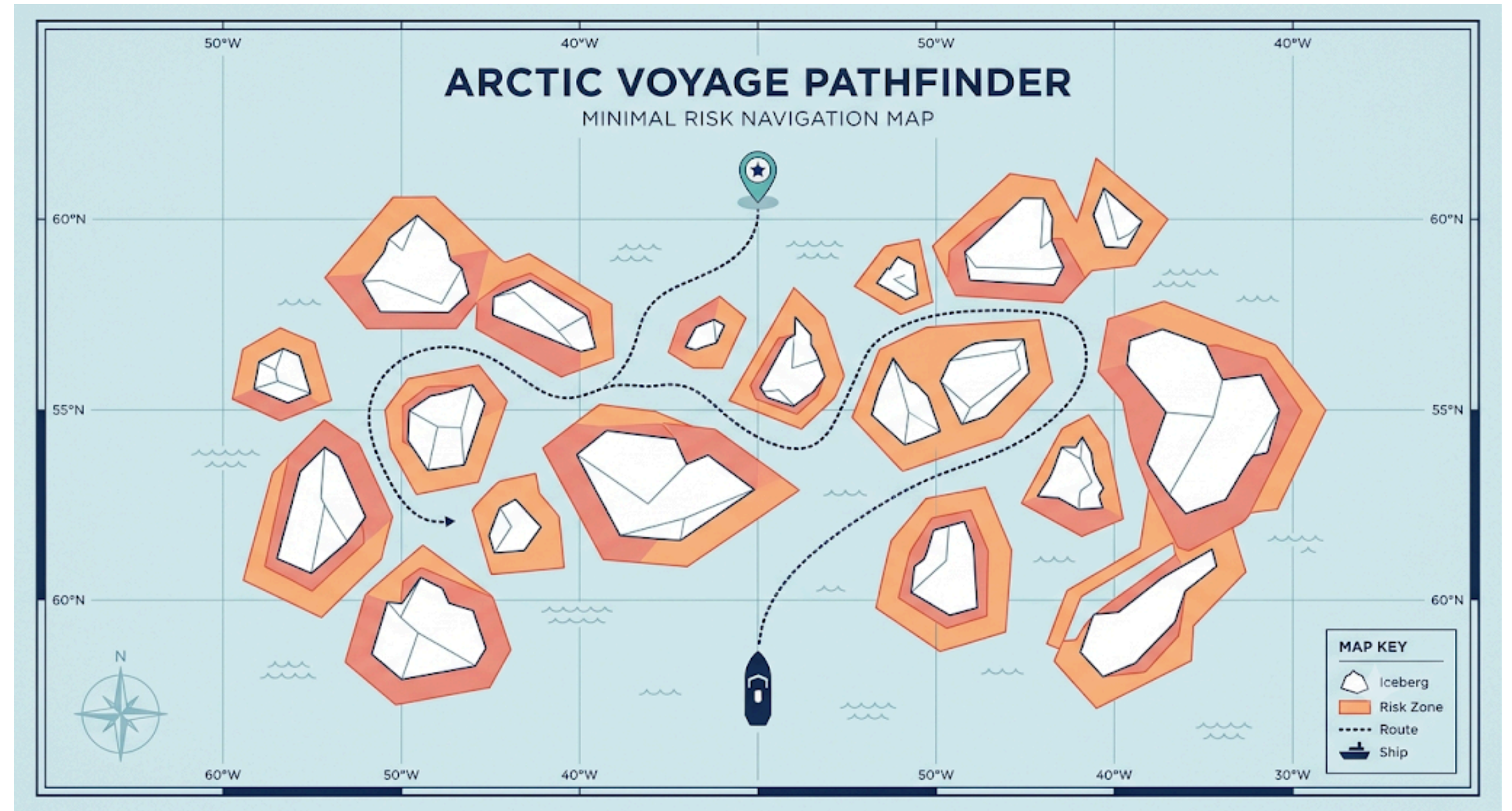
Potential challenges / risks and strategies for overcoming them

Challenge / Risk	Strategy to overcome
Unavailability of Antarctica specific data	Bootstrap with AI4Arctic (Arctic) pretraining, fine-tune on BYU/NIC Antarctic tracks; physics-residual architecture needs far less labeled data than a pure CNN
No trained models available for all the purposes	Prioritize physics-based drift as a fallback baseline for any module without a trained model yet; incrementally replace with learned residuals
Low-bandwidth polar connectivity	Onboard distilled model + delta-only sync design



Potential impact on the target audience

- Safer navigation: forward-looking risk, not a stale chart of where ice was.
- Fleet-wide value compounds : every ship's confirmed/refuted sighting improves the shared risk map for others.
- Route/voyage planning around seasonal safe windows, reducing rerouting and damage costs.
- Early warning on new icebergs before they're even fully calved.



RESEARCH AND REFERENCES

- Buus-Hinkler et al., "AI4Arctic/ASIP Sea Ice Challenge Dataset," ESSD, 2022
- Chen et al., "MMSealce: Multi-task Mapping of Sea Ice Parameters," The Cryosphere, 2024
- "IDRIFTNET: Physics-Driven Spatiotemporal Deep Learning for Iceberg Drift Forecasting," 2025 (arXiv:2507.00036)
- Statoil/C-CORE Iceberg Classifier Challenge (Kaggle)
- BYU/NIC Antarctic Iceberg Tracking Database
- Andersson et al., "Iceberg Drift Forecasting Using Machine Learning," 2019

Reference	What we're taking from it	Where it's used in our pipeline
(Uchen et al., 2012)	New dataset SAR / sighting	Training data for the sea-ice m(perception stage).
(Uchen et al., 2010)	The dataset iceberg poving fore projection	Training data for the Antarctic module
Balsing et al., 2017)	The dataset intratlior oola-ice projection traider mah	Training data for the antarctic perceptions
(Ldving et al., 2011)	The dataset assaistat in taning	Training data for the forecast
Honmin et al., 2013)	The sea-ice iceberg classifing has canden te anticipitors	Training data for the sea-ice mapping (Perception stage). Arreivious Arctic transfers
Dolnien et al., 2017)	The stagerift Iceberg poving forecasting using drift	Training data for the antarctic module (perception stage). Since it's Arctic-erntarctic scenes
Bayang et al., 2008) (2020)	The A* replan route (only if risk exceeds threshold)	Arpeats every sat pass ~6-2 hours

